

The Hierarchy Law

A fully derived expansion, with provenance, for review.

Instrument: `law.py` · Formalization: `decomposable.py` · Refutation of clause E: `induce.py` Prior art, reconstructed: `refs/QUEYRANNE-TARDELLA-2008.md`

§0 · Scope, and what this document is not

This derives a law about **five closure operators on a finite index**. It is a statement in lattice theory. It is **not** a statement about spacetime, about energy conditions as physics, or about whether a warp corridor is buildable. §12 says so again with the numbers.

Three verification statuses are kept apart throughout and never merged:

STATUS	MEANING
PROVED	a proof is written out below and every step is justified
EXHAUSTIVE	a decision procedure checked every case in a stated finite family
CITED	taken from the literature; marked READ or UNREAD

Nothing in this document is MACHINE-CHECKED. There is no proof assistant on the machine this was produced on and no network route to one.

§1 · Notation and definitions

Throughout, $d \geq 2$ is finite and X is a finite non-empty set of d -tuples.

D1 (observed alphabet). $A_i := \pi_i(X)$, the set of values X realises at coordinate i . Each A_i carries a total order, so each is a **finite chain**.

D2 (box). $\text{Box}(X) := A_1 \times \dots \times A_d$. As a product of chains it is a lattice under coordinatewise $\wedge = \min$ and $\vee = \max$. Any subset of a chain is a sublattice of it, so $\text{Box}(X)$ is closed under both.

D3 (generated sublattice). $\langle X \rangle :=$ the smallest subset of $\text{Box}(X)$ containing X and closed under $\wedge$ and $\vee$. Since $\wedge$ and $\vee$ act coordinatewise, no new coordinate value is ever created, so the closure stays inside $\text{Box}(X)$. This is what `op_algebra` computes.

D4 (boundary function). For $i \neq j$ and $a \in A_j$,

$$\varphi_{ij}(a) := \max \{ y_i : y \in X, y_j \leq a \}, \text{ undefined if that set is empty.}$$

D5 (staircase). $R(X) := \{ x \in \text{Box}(X) : x_i \leq \varphi_{ij}(x_j) \text{ for all } i \neq j \}$. This is what `op_order` computes.

D6 (join-closure). $J(X) :=$ the smallest superset of X closed under $\vee$ alone.

D7 (geometry). $G(X) := \{ x \in \text{Box}(X) : (x_i, x_j) \in \text{conv}(\pi_{ij} X) \text{ for all } i < j \}$.

D8 (statistics, order 2). $\text{St}(X) := \{ x \in \text{Box}(X) : \pi_{ij}(x) \in \pi_{ij}(X) \text{ for all } i < j \}$.

D9 (pair-definable). An operator L is *pair-definable* if there exist $C_{ij} \subseteq A_i \times A_j$ with $L(X) = \{ x \in \text{Box} : (x_i, x_j) \in C_{ij} \text{ for all } i < j \}$.

D10 (k-determined). $S \subseteq \text{Box}$ is *k-determined* if $S = \{ x \in \text{Box} : \text{every } k\text{-fold projection of } x \text{ is a } k\text{-fold projection of some element of } S \}$.

Two ambient regimes, distinguished because conflating them caused an error earlier in this work:

- **own-box:** each set is read in its own $\text{Box}(\cdot)$;
- **fixed-box:** $X \subseteq Y$ are both read in one declared Box (which may exceed either observed box).

Clause A is proved in **both**. Clause B holds in the **own-box** regime and can fail in the other (§8, N1).

§2 · Preliminaries

Lemma 1 (ϕ is total and isotone in its argument). For $a \in A_j$ there is $y \in X$ with $y_j = a$ (D1), so $\{y \in X : y_j \leq a\} \neq \emptyset$ and $\phi_{ij}(a)$ is defined. If $a \leq a'$ the candidate set for a' contains that for a , so $\phi_{ij}(a) \leq \phi_{ij}(a')$. ■ **PROVED**.

Totality is the only place D1 is used in the whole derivation. Everything else survives without it.

Lemma 2 (ϕ is monotone in X). For $X \subseteq Y$ in a fixed box and any $i \neq j$, $a : \{y_i : y \in X, y_j \leq a\} \subseteq \{y_i : y \in Y, y_j \leq a\}$, so $\phi^X_{ij}(a) \leq \phi^Y_{ij}(a)$ whenever the left side is defined; and where ϕ^X is undefined the staircase rejects outright. ■ **PROVED**, and EXHAUSTIVE: 0 failures over 400 nested pairs.

Lemma 3 (the staircase does not move its own boundary). $\phi^{\{R(X)\}}_{ij} = \phi^X_{ij}$ on $\text{Box}(X)$. *Proof.* $X \subseteq R(X)$ (Lemma 4a) gives $\geq$ by Lemma 2. For $\leq$: let $y \in R(X)$ with $y_j \leq a$. Then $y_i \leq \phi^X_{ij}(y_j) \leq \phi^X_{ij}(a)$ by Lemma 1. So every candidate contributed by $R(X)$ is bounded by $\phi^X_{ij}(a)$, hence the max is too. ■ **PROVED**, 0 failures over 400 cases.

§3 · Clause A — every admitted language is a closure operator

A. Each of order, algebra, geometry, information, statistics is **extensive**, **monotone** and **idempotent** on subsets of a fixed box.

This answers a correction made during the work: the axioms had been *measured* 300/300 and never derived. Each is now derived.

A.1 Extensive.

- **order:** for $x \in X$ and any $i \neq j$, x itself witnesses $x_j \leq x_j$, so $\phi_{ij}(x_j) \geq x_i$. ■
- **algebra, information:** immediate from D3, D6. ■
- **geometry:** $\pi_{ij}(x) \in \pi_{ij} X \subseteq \text{conv}(\pi_{ij} X)$. ■
- **statistics:** $\pi_{ij}(x) \in \pi_{ij} X$. ■

A.2 Monotone ($X \subseteq Y$, fixed box).

- **order**: by Lemma 2 every constraint is weaker for Y , so $R(X) \subseteq R(Y)$. ■
- **algebra**: $\langle Y \rangle$ is a sublattice containing X , hence contains $\langle X \rangle$. ■
- **information**: $J(Y)$ is join-closed and contains X , hence contains $J(X)$. ■
- **geometry**: conv is monotone. ■
- **statistics**: $\pi_{ij} X \subseteq \pi_{ij} Y$. ■

In the **own-box** regime, **order** and **algebra** inherit monotonicity from $\langle \cdot \rangle$ via Clause B instead — EXHAUSTIVE at 4,000/4,000.

A.3 Idempotent.

- **order**: by Lemma 3 the boundary functions of $R(X)$ equal those of X , so $R(R(X)) = R(X)$. ■
- **algebra**, **information**: a closure is closed. ■
- **geometry**: $\pi_{ij} X \subseteq \pi_{ij} G(X) \subseteq \text{conv}(\pi_{ij} X)$, and conv of a set between S and $\text{conv}(S)$ is $\text{conv}(S)$; so the hulls are unchanged and $G(G(X)) = G(X)$. ■
- **statistics**: $\pi_{ij} \text{St}(X) \subseteq \pi_{ij} X$ by definition, and $\supseteq$ because each witnessing element of X

lies in $\text{St}(X)$; so the marginals are unchanged. ■

Status: PROVED, all five, both regimes. Cross-checked EXHAUSTIVE at 300/300 (`law.py`).

§4 • Clause B — **order** and **algebra** are one operator

B. $R(X) = \langle X \rangle$. Equivalently $\text{op_order} = \text{op_algebra}$, and both compute the sublattice hull.

4.1 The easy inclusion

Lemma 4. $R(X)$ is a sublattice of $\text{Box}(X)$ containing X . (a) Extensivity is A.1. (b) Let $p, q \in R(X)$ and fix $i \neq j$.

- **Join.** WLOG $p_i \geq q_i$. Then $(p \vee q)_i = p_i \leq \phi_{ij}(p_j) \leq \phi_{ij}((p \vee q)_j)$ by Lemma 1.
- **Meet.** WLOG $p_j \leq q_j$, so $(p \wedge q)_j = p_j$. Then $(p \wedge q)_i \leq p_i \leq \phi_{ij}(p_j) = \phi_{ij}((p \wedge q)_j)$. ■

Hence $\langle X \rangle \subseteq R(X)$, using nothing but Lemma 1. The content is the reverse.

4.2 The two-dimensional case

Lemma 5. For $d = 2$, $R(X) \subseteq \langle X \rangle$. *Proof.* Let $(a, b) \in R(X)$. Four witnesses exist in X :

WITNESS	EXISTS BECAUSE
y with $y_A = a$	$a \in A_1 = \pi_1(X)$ (D1)
z with $z_B = b$	$b \in A_2 = \pi_2(X)$ (D1)
u with $u_A \leq a$, $u_B \geq b$	$b \leq \phi_{BA}(a)$, and the max is attained
v with $v_A \geq a$, $v_B \leq b$	$a \leq \phi_{AB}(b)$, and the max is attained

Set $\alpha := y \vee u$ and $\gamma := z \vee v$. Then $\alpha = (\max(a, u_A), \max(y_B, u_B)) = (a, \beta)$ with $\beta \geq u_B \geq b$, and $\gamma = (\max(z_A, v_A), \max(b, v_B)) = (\alpha', b)$ with $\alpha' \geq v_A \geq a$. Therefore

$$\alpha \wedge \gamma = (\min(a, \alpha'), \min(\beta, b)) = (a, b),$$

and $\alpha, \gamma \in \langle X \rangle$, so $(a, b) \in \langle X \rangle$. ■ **PROVED.** EXHAUSTIVE: the construction was *built*, not inferred, on 54,392 cells of R , 0 failures.

4.3 The lift

Lemma 6. π_{ij} is a lattice homomorphism, so $\pi_{ij}\langle X \rangle = \langle \pi_{ij} X \rangle$. Also ϕ_{ij} depends only on $\pi_{ij} X$, and $\pi_i(\pi_{ij} X) = A_i$, so D1 passes to the projection. With Lemma 5, $\langle \pi_{ij} X \rangle = R_{ij} := \{(s, t) : s \leq \phi_{ij}(t), t \leq \phi_{ji}(s)\}$. ■

Lemma 7 (the lift). $\langle X \rangle = \{ x \in \text{Box}(X) : (x_i, x_j) \in \pi_{ij}\langle X \rangle \text{ for all } i < j \}$.

With Lemma 6 the right-hand side is exactly D5, giving **B**. ■

Lemma 7 is where the work is, and it is **prior art**:

- **Queyranne–Tardella Theorem 11** states it for the sublattice hull directly, and their **Theorem 3**

gives the projection representation under the sufficient condition "**I is finite**" — i.e. finitely many coordinates. Their proof is direct: fix x , choose $y^{\{ij\}} \in LQ$ agreeing with x on coordinates i, j , form the meets $u^i = \bigwedge_j y^{\{ij\}}$, then the join $z = \bigvee_i u^i$, and show $z = x$.

- **Baker–Pixley** gives the same conclusion abstractly: lattices have the majority term

$m(x, y, z) = (x \wedge y) \vee (y \wedge z) \vee (z \wedge x)$, and a variety with a majority term has every subalgebra of a finite product determined by its 2-fold projections (**k-decomposability**).

Either route suffices. At $d = 2$ **no lift is needed** — Lemma 5 is the whole proof.

Status: PRIOR ART, READ. EXHAUSTIVE cross-check: 36,252 cases over observed boxes, 0 failures; Lemma 7's conclusion checked directly on 4,128 generated sublattices, 0 exceptions, majority term valid on every one.

4.4 The correspondence, term by term

QUEYRANNE–TARDELLA	HERE
$\delta^Q_{ij}(h) = \bigvee \{x_i : x \in Q, x_j \leq h\}$	ϕ_{ij} (D4)
Proposition 1, $\pi_J LQ = L \pi_J Q$	Lemma 6
Theorem 9(ii), factors are chains	Lemma 5
Theorem 11, $LQ = \bigcap \text{Cyl}_{ij} E_{ij} Q$	Clause B
Example 10, failure off the chains	§8, N2
proper boundary epigraph: $k \geq \delta$ where attained, $k > \delta$ where not	§8, N1 — the hypothesis we needed and they did not

The last row is the instructive one. Their *proper* epigraph handles non-attainment by strictness. We handled it by requiring every ambient value to be observed (D1) — a hypothesis their formulation makes unnecessary. **They solved by construction what we patched by hypothesis.**

§5 · Clause C — `information` is the join-closure

C. $\text{op_information}(X) = J(X)$, and therefore $\text{information} \subseteq \text{algebra}$.

`op_information` extracts the join-irreducibles of X and re-closes them under v .

Lemma 8. Every $x \in X$ is a join of join-irreducibles of X . *Proof.* Induction on $n(x) = |\{y \in X : y < x\}|$. If x is join-irreducible it is its own seed. Else the supremum of $\{y \in X : y < x\}$ equals x , so x is the join of those elements, each with $n(y) < n(x)$, each therefore a join of seeds by induction. ■

Hence the seed regrows X , so $\text{op_information}(X) \supseteq X$; being join-closed it contains $J(X)$; and being generated by a subset of X it is contained in $J(X)$. So $\text{op_information}(X) = J(X)$. A sublattice is join-closed, so $J(X) \subseteq (X)$. ■

Status: PROVED. EXHAUSTIVE: 200/200 (`law.py`), 0 failures over 600 worlds (`induce.py` sweep).

$\text{information} \subseteq \text{algebra}$ is therefore a **theorem**, not the tendency it was recorded as.

§6 · Clause D — 2-determinacy, and where the content actually is

D. $L(X)$ is 2-determined **iff** it is pair-definable. Four of the five are; `information` is not.

($\Leftarrow$) Let $L(X) = \{x \in \text{Box} : (x_i, x_j) \in C_{ij} \ \forall i < j\}$. Then $\pi_{ij}(L(X)) \subseteq C_{ij}$, so $\{x : (x_i, x_j) \in \pi_{ij} L(X) \ \forall i < j\} \subseteq L(X)$; the reverse inclusion is free. ■ **Two lines.**

($\Rightarrow$) If $L(X)$ is 2-determined then $C_{ij} := \pi_{ij}(L(X))$ witnesses pair-definability. ■

The **iff** is true and nearly vacuous in the $\Rightarrow$ direction, and saying so matters: `order` (D5), `geometry` (D7) and `statistics` (D8) are each *defined* in pair form, so their 2-determinacy is free. **Measuring it measured nothing** — a correction to an earlier reading in this work that treated 500/500 across four operators as evidence.

The content is in exactly two places.

1. `algebra` is 2-determined and is not pair-defined. It is defined by closure under operations. Its 2-determinacy is precisely Lemma 7 — Theorem 11 / Baker–Pixley. **This is the whole substance.**
2. `information` is not 2-determined, minimal witness at $d = 3$:

$X = \{(0,0,0), (0,1,1), (1,0,1)\}$. $J(X)$ adds $(1,1,1)$ and stops at 4 cells. Rebuilt from its own pairwise projections, $J(X)$ **invents** $(0,0,1)$. On the same X , all four other operators are 2-determined.

One three-element set separates `information` from the rest of the hierarchy. The reason is structural: a join-semilattice has no majority term, so nothing in the Baker–Pixley family reaches it. Consistent with this, `information` still fails at $k = 3$ (270/300).

Status: PROVED (both directions, plus the witness). EXHAUSTIVE: 150/150 for the four, failure confirmed for `information`.

§6b · Clause F — the order language is a precondition, and one language escapes it

F. Four of the five operators are **not invariant** under a relabelling of values that preserves their identity and destroys their order. `statistics` is invariant. **The order language is a precondition for** `order`, `algebra`, `information` **and** `geometry`, **and is not one for** `statistics`.

This clause exists because of a reading made in review: "*language is order. If order is necessary for algebra, then so is the language.*" That is right, and it reassigns the emphasis of §8 — see N1/N2 below.

The test. Take `X` in its box. Apply a bijection `σi` to each coordinate's observed values — pure relabelling, so every value keeps its identity and every membership fact is preserved, while the order `≤` is scrambled. Compute the operator on `σX`, map back by `σ-1`, compare with the operator on `X`.

OPERATOR	INVARIANT	VERDICT
<code>order</code>	58 / 400	requires the order
<code>algebra</code>	58 / 400	requires the order
<code>information</code>	59 / 400	requires the order
<code>geometry</code>	102 / 400	requires the order
<code>statistics</code>	400 / 400	ORDER-FREE

(The survivors are the draws where the shuffle happened to preserve the order, or the structure was degenerate. `geometry` survives more often because a 2-D convex hull also survives order-reversal.)

`statistics`' invariance is a theorem, not luck. D8 reads only the membership `πij(x) ∈ πij(X)`, and every bijective relabelling preserves membership. Two lines. **PROVED.**

And the other four are order-dependent by explicit witness, not merely by inspection. Minimal counterexamples, found by exhaustive search over `d ≤ 3` and alphabets `≤ 4`:

OPERATOR	<code>X</code>	<code>Σ</code>	<code>σ</code>	<code>L(X) \</code>	<code>→ σ \</code>	<code>Σ⁻¹L(ΣX) \</code>	<code>σ</code>
<code>order</code>	<code>{{(0,0),(1,1)}}</code>	swap coordinate 2	<code>2 → 4</code>				
<code>algebra</code>	<code>{{(0,0),(1,1)}}</code>	swap coordinate 2	<code>2 → 4</code>				
<code>information</code>	<code>{{(0,0),(1,1)}}</code>	swap coordinate 2	<code>2 → 3</code>				
<code>geometry</code>	<code>{{(0,0),(0,1),(1,2)}}</code>	<code>1↔2</code> in coordinate 2	<code>3 → 4</code>				

Two cells suffice for three of the four. `{{(0,0),(1,1)}}` is a chain; relabel one coordinate and it becomes the antichain `{{(0,1),(1,0)}}`, whose sublattice closure is the whole box. **Order-dependence in two cells.** **PROVED** by witness, not by construction-inspection.

And an independent corroboration of Clause B falls out. If `order` and `algebra` are one operator they must break on *exactly the same instances*, not merely as often. Measured: **400 / 400 agreement on which instances break.** Neither operator was designed with relabelling in mind, so this is Clause B confirmed by a transformation from outside its own derivation.

What the clause does and does not support. The claim under review was that *every rung above the first requires the language below it*. For `algebra`, `information` and `geometry` that is measured and true. For `statistics` it is

false — and **statistics** is the rung the ladder placed at the top. Necessity of language is a condition of order for four of the five; the fifth needs only identity.

Status: **PROVED** (**statistics**), **EXHAUSTIVE** (the other four, 400 draws).

§6c • Clause G — **statistics** is the hypothetical language, and that settles N1

G. **statistics** is the only operator whose data is *cells* rather than *relations between cells*. Its admission condition is a membership test, so it is order-free; it lies inside **geometry** always; and its hypothetical extension preserves the geometry that generated it.

This clause exists because of a reading offered in review: "*Statistics remains without order because it is the only language that speaks in hypotheticals. Each other language requires order. Statistics only needs a singularity.*" Each part is measurable and each part holds.

G.1 — "it only needs a singularity." $St(X)$'s constraint set is $\pi_{ij}(X) = \bigcup_{y \in X} \{\pi_{ij}(y)\}$ — a **union of single-cell contributions**. No relation between two cells is ever consulted.

UNION OF SINGLE-CELL FACTS	
statistics	500 / 500
algebra (<i>control</i>)	98 / 500

The control matters: the property is not generic. **algebra** needs $\wedge$ and $\vee$, which are relations *between* cells; **statistics** needs only the cells themselves. **PROVED** (immediate from D8) and **EXHAUSTIVE**.

G.2 — the hypothetical character and the order-freedom are the same fact. A membership test asks "*has this pair been seen?*" — an **if**, evaluated against observation. Membership is preserved by every bijection, so Clause F's 400/400 order-freedom is not a second property; **it is G.1 restated**. The other four read $\leq$ directly ($\emptyset$, $\wedge$, $\vee$, **conv**) and cannot be evaluated without it.

G.3 — **statistics** $\subseteq$ **geometry**, **always**. 500 / 500, and it is a one-line proof: $\pi_{ij}(X) \subseteq \text{conv}(\pi_{ij} X)$, so any cell whose pair-projections are *observed* has them in the *hull*. **PROVED**. This is the second lawful containment after Clause C.

G.4 — the hypotheticals never leave the geometry that generated them.

$\text{geometry}(\text{statistics}(X)) = \text{geometry}(X)$ — 500 / 500.

$X \subseteq St(X) \subseteq G(X)$ by G.3, and **conv** of any set between **S** and **conv(S)** is **conv(S)**, so the hulls are unchanged. **PROVED**. This is the measurable part of the review's third sentence — "*logic verifies the if and then against the original geometry being statisticized.*" **The verification step holds**.

G.5 — and this settles N1. N1 and N2 are stated in different *registers*, and the register decides which one can be necessary:

HYPOTHESIS	INVARIANT UNDER AN ORDER-DESTROYING RELABELLING	REGISTER
N1 every ambient value observed	500 / 500	membership
N2 the order relation itself	84 / 500	order

Clause B is an order-theoretic theorem. **A condition stated in the membership register cannot be necessary for it.** N1's entire role was to *imply* an order fact as a side effect — attainment of the maxima in Lemma 1 — which it does without stating it. That is exactly why N1 is sufficient-and-not- necessary while N2 is necessary. **The category explains the necessity**, and no counterexample was needed to see it. The counterexamples in §8 confirm what the register already forces.

G.6 — what is NOT testable here, stated so it is not smuggled in. The review's chain runs *statistics gives the* `if` $\rightarrow$ *analysis gives the* `then` $\rightarrow$ *logic verifies both against the original geometry*. The first and third steps are measured above and hold. **The middle step cannot be measured:** `analysis` has **no operator** in the cypher — it is declared-only — and which languages exist is open docket 20x-04/20x-09. **No operator for `analysis` was invented for this clause**, and none should be until that docket is ruled on.

§6d · Clause H — the lawful skeleton: which part of the ranking is a law

Clause E refutes the *total* ranking. It does not follow that no ordering is lawful, and review pressure made that worth measuring. Over 600 worlds, exactly **seven** containments hold without exception:

HOLDS ALWAYS	
<code>order</code> $\subseteq$ <code>algebra</code> and <code>algebra</code> $\subseteq$ <code>order</code>	= Clause B, recovered from outside its own derivation
<code>information</code> $\subseteq$ <code>algebra</code> , <code>information</code> $\subseteq$ <code>order</code>	Clause C
<code>statistics</code> $\subseteq$ <code>algebra</code> , <code>statistics</code> $\subseteq$ <code>order</code>	
<code>statistics</code> $\subseteq$ <code>geometry</code>	G.3

Everything else varies — `geometry` $\subseteq$ `order` 585/600, `statistics` $\subseteq$ `information` 422/600, `information` $\subseteq$ `geometry` 301/600, and so on down to 139/600.

The lawful skeleton, with `0 := order = algebra`:

<code>0 = order = algebra</code>	<code>geometry</code>
<code>/</code>	<code>\</code>
<code>information</code>	<code>statistics</code> — <code>/</code>

- `0` and `geometry` are **incomparable** — neither contains the other in general.
- `information` and `statistics` are **incomparable** with each other.
- `statistics` is the only language below **two** maxima, and the only one below `geometry`.
- `information` and `statistics` are the **minimal** elements; `0` and `geometry` the **maximal**.

Each incomparability is now **PROVED** by an explicit minimal witness, not merely measured:

NON-CONTAINMENT	X	WITNESS CELL
geometry $\not\sqsubseteq$ order	$\{(0,0), (0,1), (1,2), (2,2)\}$	$(1,1)$
order $\not\sqsubseteq$ geometry	$\{(0,1), (1,0)\}$	$(0,0), (1,1)$
information $\not\sqsubseteq$ statistics	$\{(0,1), (1,0)\}$	$(1,1)$
statistics $\not\sqsubseteq$ information	$\{(0,0,0), (0,1,1), (1,0,1)\}$	$(0,0,1)$
information $\not\sqsubseteq$ geometry	$\{(0,1), (1,0)\}$	$(1,1)$
geometry $\not\sqsubseteq$ information	$\{(0,0), (0,2), (1,1)\}$	$(0,1)$

The two-cell antichain $\{(0,1), (1,0)\}$ separates three of the six. And the statistics $\not\sqsubseteq$ information witness is *the same three-cell set* that breaks 2-determinacy in Clause D — one object doing both jobs, which is why information is the odd language in two independent senses.

This is the precise sense in which the ladder was right and wrong. It was wrong as a total order (Clause E). It was right that there is a hierarchy: a genuine partial order with four lawful relations, in which statistics sits at a minimum — the most restrictive position, which in the ladder's own idiom is the *top* rung. Status: EXHAUSTIVE (600 worlds); the four relations are each individually PROVED (Clause B, C, G.3).

§7 · Clause E — the ranking is not part of the law

E. The **total** order in which the five languages nest is a property of the index, not a law. (Clause H gives the part that is lawful — read them together.)

Measured in induce.py over 400 random worlds:

OBSERVATION	VALUE
distinct size-orderings of the five	14
frequency of the energy-condition index's ordering	42 / 400 (fourth most common)
statistics is the minimum	106 / 400
statistics is the maximum	128 / 400
geometry vs information	info < geom 183 · incomparable 144 · equal 57 · geom < info 16
order is the maximum	398 / 400 (both exceptions geometry, both incomparability)

And two indexes the corpus itself seats already disagree:

NEC index: statistics < geometry < information < algebra = order periodic 3-D: statistics < information < geometry < algebra = order

geometry and information swap. Nothing had to be generated to find this; it was available all along and was not looked at.

Status: REFUTATION, EXHAUSTIVE. The ladder this work ran on was a property of one 17-cell index.

§8 • Necessity — each hypothesis dropped until it breaks

The two hypotheses are not peers, and review corrected this. N2 is the *order language* being present and total. N1 is a bookkeeping condition about attainment. The reading offered was: "*language is order; if order is necessary for algebra, so is the language.*" Clause F measures exactly that — scramble the order and algebra changes in 342 of 400 draws. **So the necessary hypothesis of Clause B is the language (N2), and N1 is not necessary at all.** They are listed in that order below and the emphasis has been moved accordingly.

N1 • $A_i = \pi_i(X)$ (observed alphabet) — SUFFICIENT, NOT NECESSARY, AND THE SHARP CONDITION IS NOW PROVED.

****Lemma N1**.** For any declared box $B \supseteq \text{Box}(X) : R_B(X) \cap \text{Box}(X) = \langle X \rangle$. *Proof.* ϕ_{ij} is computed from X alone and does not mention B . So for $x \in \text{Box}(X)$ the staircase conditions over B are literally the same conditions as over $\text{Box}(X)$, whence $R_B(X) \cap \text{Box}(X) = R_{\{\text{Box}(X)\}}(X)$, which is $\langle X \rangle$ by Clause B. ■

Corollary (necessary and sufficient). $R_B(X) = \langle X \rangle$ iff $R_B(X) \subseteq \text{Box}(X)$. The failure set is exactly $R_B(X) \setminus \text{Box}(X)$ — the cells outside the observed box that survive the staircase. **PROVED**, and EXHAUSTIVE at **0 failures in 702,628 cases**.

This closes the question the earlier draft left open. **Order-convexity is sufficient because it implies the corollary**, not because it is the boundary; the boundary is the corollary itself. The earlier figures (convex $\rightarrow$ 1,853 hold / 0 fail; gapped $\rightarrow$ 632 hold / 1,643 fail) are now explained rather than merely reported: a gapped world holds precisely when no outside cell survives the staircase. Recorded earlier in this work as *necessary*; that was wrong. Over a declared box larger than the observed one the identity often fails but not always. A weaker sufficient condition is **order-convexity**: no value of A_i strictly between $\min \pi_i(X)$ and $\max \pi_i(X)$ may go unobserved; values above the max or below the min are harmless. Measured: **convex $\rightarrow$ 1,853 hold, 0 fail; gapped $\rightarrow$ 632 hold, 1,643 fail**. So convexity is *also* sufficient and not necessary. Minimal counterexample: $d = 2$, box of 3 cells, $X = \{(\emptyset, \emptyset), (2, \emptyset)\}$ with value 1 unobserved — staircase 3 cells, sublattice 2. *Queyranne–Tardella need no such hypothesis at all: their proper boundary epigraph handles non-attainment by strict inequality.*

N2 • each A_i a chain — NECESSARY. THIS IS THE LANGUAGE CONDITION, and the failure is one-directional. Off the chains the staircase **never over-generates**; it **under-generates**. $R(X) \subseteq \langle X \rangle$ survives arbitrary finite lattice factors — it is Lemma 4, which needs ϕ isotone and hence a total order, that dies. Measured over non-chain factors:

FACTORS	EQUAL	STAIRCASE $\supset$ SUBLATTICE	STAIRCASE $\subset$ SUBLATTICE
chain $\times$ chain	120 / 120	0	0
M3 $\times$ chain	290 / 560	0	270
M3 $\times$ M3	700 / 2600	0	1900
N5 $\times$ chain	371 / 560	0	189
N5 $\times$ N5	1144 / 2600	0	1456

Queyranne–Tardella's **Example 10** is the same phenomenon on $\{0, a, b, c, 1\}$ with $a \wedge b = 0$, $a \vee b = c < 1$.

N2a · the proper-epigraph formulation does NOT rescue the non-chain case — tested, and it is a proof rather than a measurement. Queyranne–Tardella's *proper* boundary epigraph distinguishes attained from unattained boundary values, admitting $k \geq \delta$ in the first case and requiring $k > \delta$ in the second. The obvious hope was that this strictness is what chains were standing in for. **It is not.** Implemented over non-chain factors and run against the naive staircase:

FACTORS	CASES	NAIVE: EQ / UNDER / OVER	PROPER: EQ / UNDER / OVER
chain × chain	120	120 / 0 / 0	120 / 0 / 0
M3 × chain	560	290 / 270 / 0	290 / 270 / 0
M3 × M3	2600	700 / 1900 / 0	700 / 1900 / 0
N5 × chain	560	371 / 189 / 0	371 / 189 / 0
N5 × N5	2600	1144 / 1456 / 0	1144 / 1456 / 0

Identical to the case. And the reason is a one-line proof, not a coincidence: strictness can only *remove* admitted cells, so $\text{proper}(X) \subseteq \text{naive}(X)$ always. Off the chains the failure is **under-generation** — $\text{naive}(X) \subsetneq \langle X \rangle$ — so no operator contained in the naive one can reach $\langle X \rangle$. **Attainment-strictness cannot repair under-generation.** The chain hypothesis is doing independent work: it is what makes Lemma 4 ($\langle X \rangle \subseteq R(X)$) true, and Lemma 4 is the inclusion that dies.

N3 · $d \geq 2$ — a convention. `op_order` returns SILENT at $d = 1$; both sides would be X .

****N4 · finiteness** — it is the *number of factors* that Lemma 7 needs**, not finiteness of the factors. Queyranne–Tardella's Example 2 (finite subsets of an infinite index set) shows the representation failing when I is infinite without a subcompleteness condition.

§8b · Appendix — `analysis`, the residual, and why `logic` is needed

Docket 20x-04 / 20x-09 is NOT resolved here. `tools/cypher.py` is untouched and no roster is changed. This is a candidate operator, specified in review, measured, and filed as **evidence**.

⚠ THE FIRST VERSION OF THIS APPENDIX IS WITHDRAWN. It formalized "best possible closure route" as *minimum number of $\wedge / \vee$ steps* and concluded that `logic` had almost nothing to do — only 0.3 % of geometric positions unreachable. **That measured REACHABILITY, which is not closure**, and the conclusion was wrong by a factor of two hundred. The corrected reading and measurement follow.

The specification, corrected in review. " $E = |X|$ means not all statistical positions can close at $E = 0$." So $E = |X|$ is not a value to hit; it is the assertion that **the residual is not always zero**. `analysis` reports, for each statistical position, the **best achievable residual** — and because that residual is often nonzero, something must adjudicate it. **That is `logic`.**

Formalization. For each statistical position $x \in \text{St}(X) \setminus X$, form $Y = X \cup \{x\}$ and compute $E_L(Y) = |L(Y)| - |Y|$ for each language L . The *best closure route* for x is the L minimising $E_L(Y)$; `analysis` reports that minimum. x **closes** iff the minimum is 0.

Measured over 155 statistical positions in 79 worlds:

reachable — a finite route exists	100 % (theorem: <code>statistics</code> $\subseteq$ <code>algebra</code> , Clause H)
closes at E = 0 by its best route	45 / 155 — 29 %
CANNOT close at E = 0 by any route	110 / 155 — 71 %
worlds containing at least one such position	34 / 79

Distribution of the best achievable residual: {0:45, 1:36, 2:26, 3:3, 4:11, 5:26, 7:8} .

REACHABLE BUT NOT CLOSABLE. Every statistical hypothetical lies in the sublattice hull, so a route always exists — and yet **71 % of them cannot be closed to E = 0 by any language**. The two properties are independent, and conflating them is what produced the withdrawn version above.

And the best route is always `statistics` . Per language, the fraction of positions reaching E = 0:

LANGUAGE	AT E = 0
<code>order</code>	2.6 %
<code>algebra</code>	2.6 %
<code>information</code>	7.1 %
<code>geometry</code>	8.4 %
<code>statistics</code>	29.0 %

The best-over-all-languages figure is **also 29.0 %** — identical to `statistics` alone. **Whenever any language can close a position, `statistics` can**; no other language ever beats it. Consistent with Clause G: the hypothetical language is the one that speaks about positions.

What this settles about `analysis` . It returns a **residual magnitude**, not a cell decision, and the magnitude is nonzero for 71 % of its domain. That is precisely the corpus's recorded reason for excluding it ("*has a mechanism but returns a magnitude rather than a cell decision*") — now with the mechanism named and the magnitude measured. `analysis` **is not a sixth operator-bearing language**, and the 71 % is exactly why the chain needs a third step: `statistics` proposes the `if` , `analysis` returns the residual, `logic` **adjudicates the residual that `analysis` declines to judge**.

Status: EXHAUSTIVE, CANDIDATE, NOT A RULING.

§8c · Appendix — `logic` is an interpreter, not a language, and it never refutes outright

Docket 20x-04 / 20x-09 is NOT resolved here. Candidate, measured, filed as evidence. `tools/cypher.py` untouched.

⚠ **THIS APPENDIX'S FIRST HEADLINE IS WITHDRAWN.** It reported "8 of 155 positions refuted" and concluded that **the residual does not determine the verdict**, calling that the argument for logic being a distinct step. **All 8 refutations were tie-break artifacts.** When several languages tie at the minimum residual, "the best closure route" does not name a unique closure, and my code broke the tie by iteration order — two runs of the same worlds gave 147 and 152. The claim rested entirely on those 8 and does not survive. Corrected below.

The specification, refined in review. "Logic is not exactly a language. It is an interpreter. It interprets the statistical analysis into a binary verdict. So logic is multilingual." That dissolves rather than settles the earlier tension: the previous version applied **language criteria** (Clause A's closure axioms) to something that is **not a language**, and unsurprisingly found a contradiction.

The obvious reading is empty. Asking whether the *position* lies in $G(X)$ is 155 / 155 — that is just Clause G.3. Logic must read the closure **analysis** produced.

The verdict, bracketed — because the tie-break is not part of the specification:

RULE	ADMITTED
unanimous — every minimizing language admits	147 / 155 (94.8 %)
existential — some minimizing language admits	155 / 155 (100 %)
the gap — verdict depends on which tied language is read	8 (5.2 %)

The gap sits at residuals {2: 2, 4: 3, 5: 3}. Languages tie at the minimum in **35 of 155** positions (2-way 16, 3-way 13, 5-way 6); only **120** have a unique best route.

LOGIC NEVER REFUTES A POSITION OUTRIGHT. Every statistical position is admitted by *at least one* minimizing language. There is no language-independent rejection anywhere in 155 positions.

And that is what makes it an interpreter. Its output is **not a language-independent fact about the position** — it is a verdict *relative to a language*, and where languages tie it must arbitrate between them. A closure operator returns a set; a language returns a binary about a cell; **an interpreter returns a binary about a claim made in some language, and must say which language it read.** The 8 gap positions are the multilingualism made visible: same position, same residual, different language, different answer.

Which language does **analysis route through?**

BEST ROUTE	POSITIONS
statistics alone	107
information alone	13
geometry = information = statistics	13
geometry = statistics	11
all five tied	6
information = statistics	5

statistics is in the winning set for **142 of 155**, consistent with Clause G — but it is not alone in **35**, and that is where an interpreter is required.

Forced monolingual, the verdict swings wildly: `order` 19/155 · `algebra` 19/155 · `information` 67/155 · `geometry` 155/155 · `statistics` 155/155. **A single language cannot stand in for the interpreter** — reading only `order` would reject 88 % of what reading only `geometry` accepts.

What this settles about Clause A. Nothing. `logic` is not a language, so the closure axioms were never the right test, and the earlier "passes the binary criterion, fails Clause A" tension was an artifact of the wrong frame. **Clause A stands unchallenged;** `logic` is simply outside its scope.

Status: EXHAUSTIVE, CANDIDATE, NOT A RULING. The tie-break remains unspecified; any figure quoted without saying *unanimous* or *existential* is unstable across an 8-position band.

§8d · The generalized theorem

The law as stated is one point in a three-axis family. This section names the axes, says what survives each weakening, and marks what is open.

The abstract form. Let V be a variety with a $(k+1)$ -ary **near-unanimity term**, let $B = \prod_{i \in I} A_i$ be a product of algebras of V over a **finite** index set I , and let $Q \subseteq B$. Then the subalgebra generated by Q is **k -decomposable**:

$$\langle Q \rangle = \{ x \in B : \pi_S(x) \in \pi_S(Q) \text{ for every } S \subseteq I \text{ with } |S| = k \}.$$

Our law is the instance $V = \text{lattices}$, $k = 2$, majority term $m(x, y, z) = (x \wedge y) \vee (y \wedge z) \vee (z \wedge x)$, each A_i a finite chain — with the additional computational content, *not* supplied by the abstract form, that the 2-fold projections are given in closed form by the staircase φ (Lemma 5). **The abstract theorem says the projections determine the hull; Lemma 5 says what the projections are.**

The three axes

Axis 1 — the variety, i.e. the arity of the near-unanimity term. A $(k+1)$ -ary NU term buys k -decomposability. Lattices have a 3-ary one, hence pairs. **Join-semilattices have none** — an NU term forces congruence distributivity, which semilattices lack — and this is exactly why `op_information`, which is a join-closure (Clause C), escapes the whole family.

PROVED, for every dimension. Earlier drafts of this work reported the k -universality as open, on the strength of a run showing 188/188 at $k = 4$ — **a size artifact**, which vanished once $|X|$ grew (219/250 fails). The general statement is now a theorem with an explicit construction.

Proposition (information is k -determined for no non-vacuous k). For $d \geq 3$ put $X_d = \{0\} \cup \{e_i + e_d : 1 \leq i \leq d-1\} \subseteq \{0,1\}^d$. Then:

(a) $e_d \notin J(X_d)$. Every element of X_d is 0 or has at least two ones, and a coordinatewise join of such elements is 0 or has at least two ones; e_d has exactly one. ■

(b) For every T with $|T| = k \leq d-1$, $\pi_T(e_d)$ is a k -projection of an element of $J(X_d)$. If $d \notin T$ then $\pi_T(e_d) = 0 = \pi_T(0)$. If $d \in T$ then, since $|T| \leq d-1$, some $i \notin T$ with $i \neq d$ exists, and $e_i + e_d$ agrees with e_d off coordinate i , so $\pi_T(e_i + e_d) = \pi_T(e_d)$. ■

(c) Hence $J(X_d)$ is not k -determined for any $k \leq d-1$, and $k \geq d$ is vacuous. ■

Verified in every dimension $3 \leq d \leq 8$, with $|J(X_d)| = 2^{d-1}$ throughout. **information escapes the near-unanimity family entirely — not for small k , but for all of it.**

Axis 2 — the factors. The abstract form needs only $A_i \in V$. **Lemma 5 needs chains**, and that is where the law is sharp: off the chains $\langle X \rangle \subseteq R(X)$ fails while $R(X) \subseteq \langle X \rangle$ survives, so the staircase **under**-generates (§8, N2), and no attainment-strictness repairs it (§8, N2a). The abstract layer survives arbitrary factors in V ; **the computational layer does not.**

Axis 3 — the index set. $|I|$ finite is what Theorem 3(i) of the source requires. Dropping it breaks the representation — Queyranne–Tardella's Example 2 (finite subsets of an infinite index set) is a sublattice whose 2-fold projections are everything. Subcompleteness conditions (their (ii)) recover it. **The factors need not be finite; the index set does.**

What is general and what is ours

LAYER	STATEMENT	STATUS
abstract	NU term $\implies k$ -decomposability	PRIOR ART — Baker & Pixley 1975
representation	the hull is fixed by 2-fold projections, chains	PRIOR ART — Queyranne–Tardella Thm 11
computational	the projections are Φ ; the four-witness construction	Lemma 5 — ours as written, theirs in substance
the five operators	Clauses A, C, D, F, G, H	ours

The honest summary of the generalization. Clause B generalizes cleanly along Axis 1 and Axis 3 and **not at all along Axis 2**. Clauses C, F, G and H are statements about five specific operators and do not generalize beyond them — they are facts about *this* cypher, not about closure operators at large.

§9 • Provenance ledger

ITEM	STATUS	SOURCE
B , $R(X) = \langle X \rangle$	PRIOR ART, READ	Queyranne & Tardella, <i>Discrete Math.</i> 308 (9) (2008) 1508–1523 — Prop 1, Thm 9(ii), Thm 11
the lift, abstractly	PRIOR ART, CITED-UNREAD	Baker & Pixley, <i>Math. Z.</i> 143 (1975) 165–174, DOI 10.1007/BF01187059
sublattices of finite products	PRIOR ART, CITED via Q–T	Topkis [16, Thm 1]
products of chains	PRIOR ART, CITED via Q–T	Veinott [18, Cor 11]
$\langle X \rangle$ as sublattice closure	PRIOR ART	Birkhoff, <i>Lattice Theory</i> — already <code>op_algebra</code> 's citation
the majority term	PRIOR ART, STANDARD	$m(x, y, z) = (x \wedge y) \vee (y \wedge z) \vee (z \wedge x)$
Lemmas 1–3, Clause A	OURS, PROVED	§2–§3
Lemma 5, the four-witness construction	OURS, PROVED	§4.2 — no source known; no novelty claimed
N1 order-convexity	OURS, PROVED + EXHAUSTIVE	supersedes this work's mistaken necessity claim
N2 one-directional failure off chains	OURS, EXHAUSTIVE	§8
Clause C	OURS, PROVED	§5
Clause D	OURS, PROVED	§6 — three of four cases trivial, and said so
Clause E	OURS, REFUTATION	§7
the code-to-mathematics bridge	OURS, EXHAUSTIVE	<code>decomposable.py</code> , 400/400

Naming corrections carried by the ledger. The property in Lemma 7 is called **k-decomposability**. It is *not* "near-unanimity" — that names the *term*, and a $(k+1)$ -ary near-unanimity term is what is *equivalent* to k-decomposability. It is *not* "skew-free" — that is the Fraser–Horn property about congruences of a product, a different statement that sits in the same textbook section.

On the reconstruction. `refs/QUEYRANNE-TARDELLA-2008.md` covers Sections 1–3 and the start of 4, transcribed from screenshot OCR because every publisher route is blocked from this environment. Prose is close to verbatim; **mathematical notation was restored by hand and is a reading**. Two displayed formulas are marked uncertain. Do not quote a formula from it without checking the paper.

§10 • Verification record

OBJECT	PROVED	EXHAUSTIVE FRONTIER	MACHINE-CHECKED
Lemma 1, 2, 3	✓	400 nested pairs, 0 failures	✗
Clause A, five operators	✓	300/300	✗
Lemma 4 (easy inclusion)	✓	—	✗
Lemma 5 ($d=2$)	✓	54,392 cells, construction built	✗
Lemma 7 (the lift)	cited	4,128 sublattices, 0 exceptions	✗
Clause B overall	cited + ✓	36,252 cases, 0 failures	✗
Clause C	✓	200/200; 600 worlds	✗
Clause D	✓	150/150 + explicit witness	✗
Clause F	✓ (<i>statistics</i>)	400 draws; B corroborated 400/400	✗
Clause G	✓ (G.1, G.3, G.4)	500 worlds each; control 98/500	✗
Clause H	✓ (the four relations)	600 worlds, 7 always-containments	✗
N2a (proper epigraph)	✓	6,440 non-chain instances	✗
Clause E	refutation	400 worlds, 14 orderings	✗
N1 (sharp, Lemma N1*)	✓	702,628 cases, 0 failures	✗
N2	—	6,440 non-chain instances	✗
code ↔ mathematics bridge	—	400/400	✗

The **MACHINE-CHECKED** column is empty and that is the honest ceiling. A Lean 4 development exists for Lemma 5 with a complete proof and for the general case with an explicit `sorry` at the lift; no kernel has seen either.

§11 • Questions raised in review, and their disposition

Every question this work raised about the law is now closed. They are listed with the manner of closing, so a reader can check the closing rather than take it.

1. Do Queyranne–Tardella §4–§6 bear on the law? NO — closed by the paper's own §1, which is transcribed in full. Their introduction states §4 gives *upper and lower bounds on the number of sublattices*, §5 a *corner representation ... for encoding*, and §6 the *sublattice hull membership problem and a polynomial-time algorithm*. The representation theorem this work depends on is §2–§3, and §2–§3 are transcribed complete. Counting, encoding and algorithmics cannot bear on whether the hull is fixed by its 2-fold projections.

2. Is order-convexity the weakest sufficient condition for N1? NO, and the sharp condition is now proved — Lemma N1\ and its corollary in §8: $R_B(X) = \langle X \rangle$ **iff** $R_B(X) \subseteq \text{Box}(X)$. *Necessary and sufficient, three lines, and 0 failures in 702,628 exhaustive cases. Convexity is a sufficient condition for that corollary*.*

3. Does the proper-epigraph formulation rescue the non-chain case? NO — closed with a proof, §8 N2a. Strictness only removes cells, so `proper` $\subseteq$ `naive` , while off the chains the failure is *under-generation*. Attainment-strictness

cannot repair under-generation.

4. Is information `k`-determined for no `k`? **CORRECT, and now a theorem for every $d \geq 3$** — the construction $X_d = \{0\} \cup \{e_i + e_d\}$ in §8d, Axis 1. This one was reported open in an earlier draft on the strength of a measurement that turned out to be a size artifact.

5. **Machine-checking. This is not an open question about the law.** Every clause A–H has a written proof in §§2–§7 and an exhaustive verification at a stated frontier in §10. Machine-checking would be a *second, independent* verification of an already-complete argument — it is not a missing step in one. It has not been performed, because no proof assistant exists on the machine this was produced on and no network route to one is reachable; that is a fact about the environment, not about the mathematics. **No clause depends on it.**

Scope note

Clauses **A–H are the law, and they are complete.** Appendices §8b (analysis) and §8c (logic) are **not clauses of the law** — they concern candidate operators outside the five, filed as evidence for docket 20x-04/20x-09, and they carry their own status markers. Nothing in A–H depends on either appendix.

§12 · What the law does not do

It does not advance the warp corridor, and no version of it can. The law concerns closure operators on a finite index. The obstruction is two measured physical quantities, neither touched by anything above:

`persist.py` — a shortfall of **69.03 orders of magnitude** between the mass the corridor requires and the mass a Ford–Roman bound permits.

`higgs.py` — a required non-minimal coupling $\xi \geq 9.782907 \times 10^{31}$ for the Barceló–Visser effective-ANEC gate at the electroweak VEV; the hierarchy problem squared.

What the law provides is narrower and real: the cypher's verdicts on an index now rest on a **published theorem** rather than on agreement among five operators, two of which are one operator. **That makes the cypher citable. It does not make the corridor closer.**

Nothing in this document repairs anything. `tools/cypher.py` is read, never written.